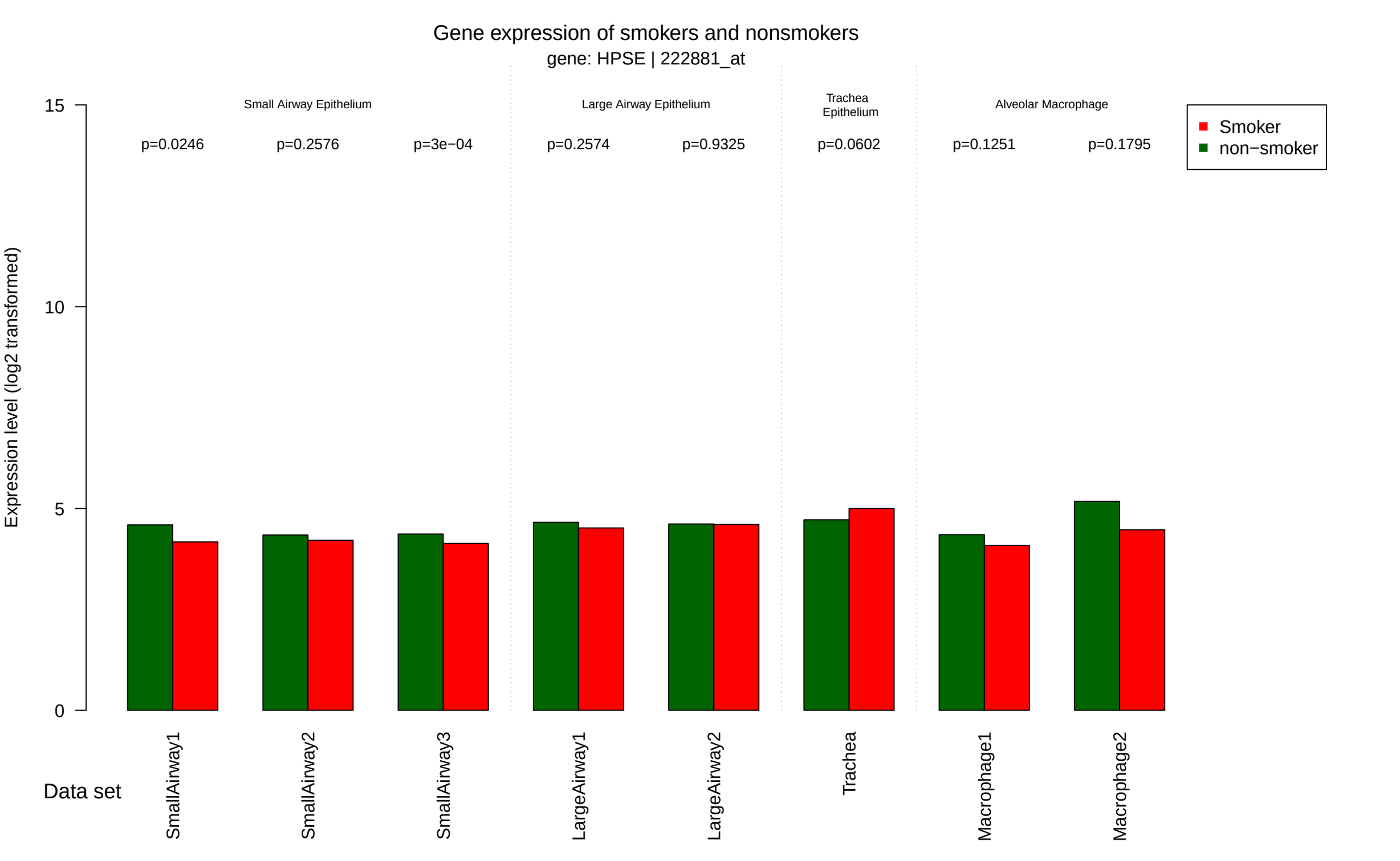
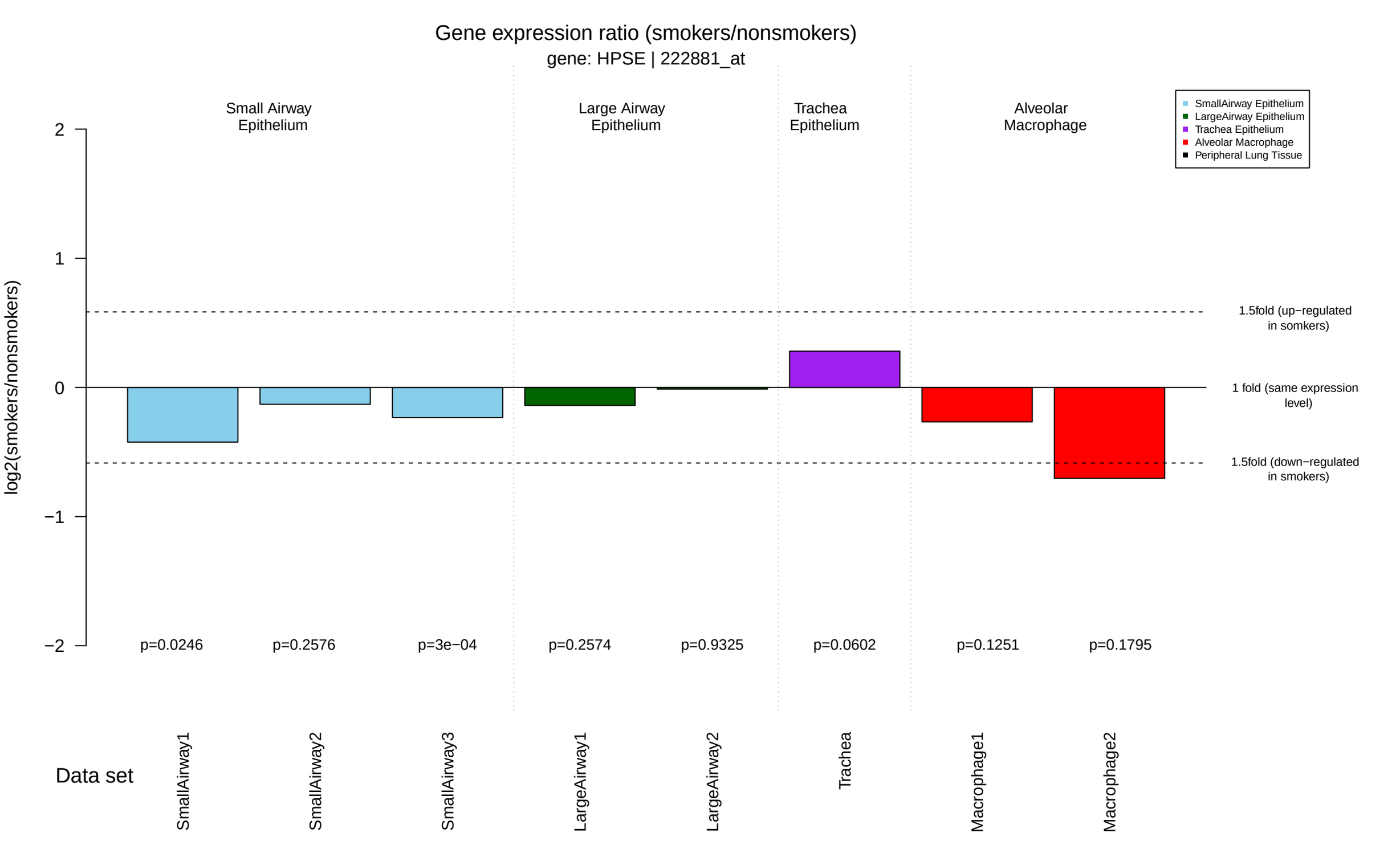
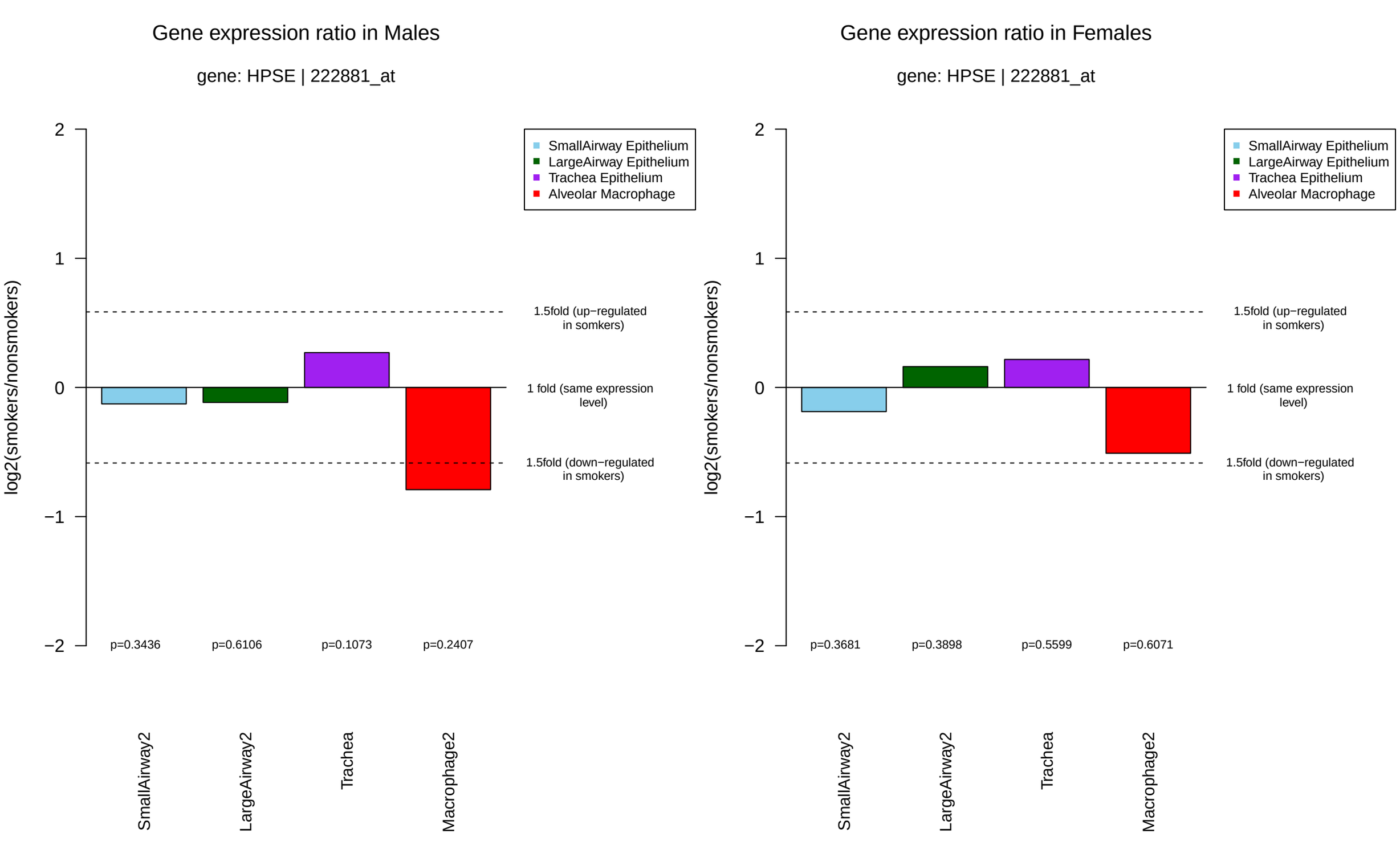
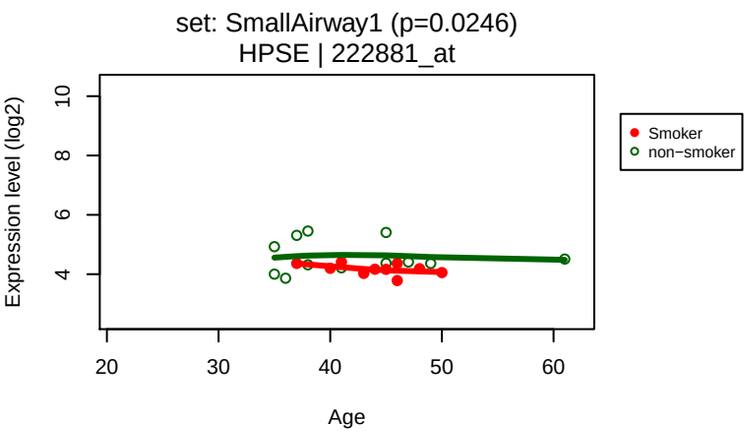
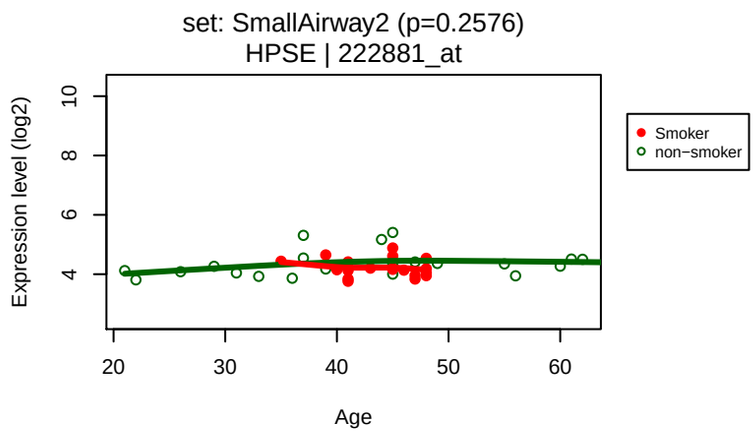
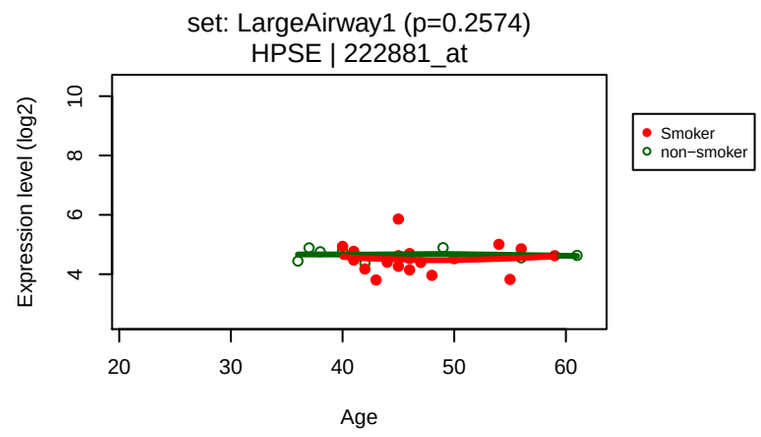

Gene expression of smokers and nonsmokers
gene: HPSE | 222881_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0246
p=0.2576
p=3e−04
p=0.2574
p=0.9325
p=0.0602
p=0.1251
p=0.1795
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

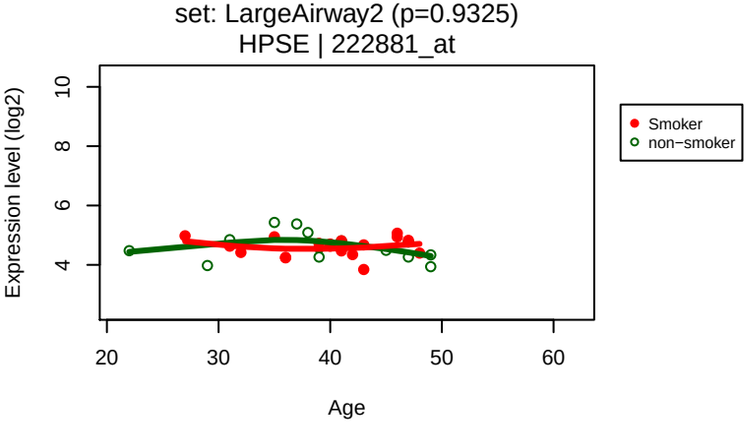

Gene expression ratio (smokers/nonsmokers)
gene: HPSE | 222881_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0246
p=0.2576
p=3e−04
p=0.2574
p=0.9325
p=0.0602
p=0.1251
p=0.1795
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

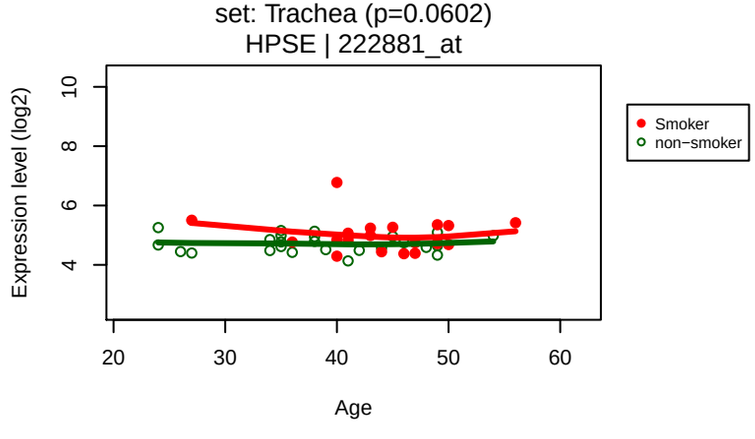

Gene expression ratio in Males
gene: HPSE | 222881_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3436
p=0.6106
p=0.1073
p=0.2407
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HPSE | 222881_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3681
p=0.3898
p=0.5599
p=0.6071
SmallAirway2
LargeAirway2
Trachea
Macrophage2

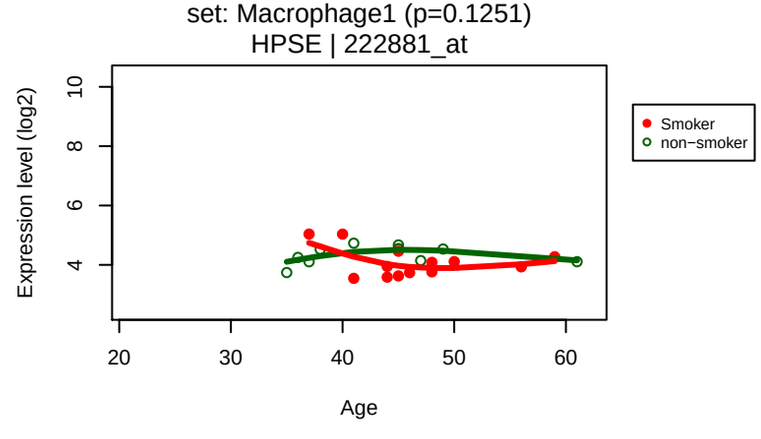

set: SmallAirway1 (p=0.0246)
HPSE | 222881_at
Expression level (log2)
Age
Smoker
non−smoker

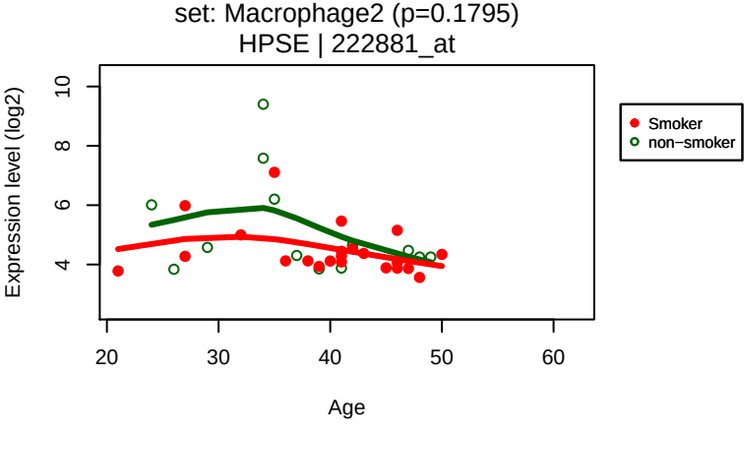

set: SmallAirway2 (p=0.2576)
HPSE | 222881_at
Expression level (log2)
Age
Smoker
non−smoker

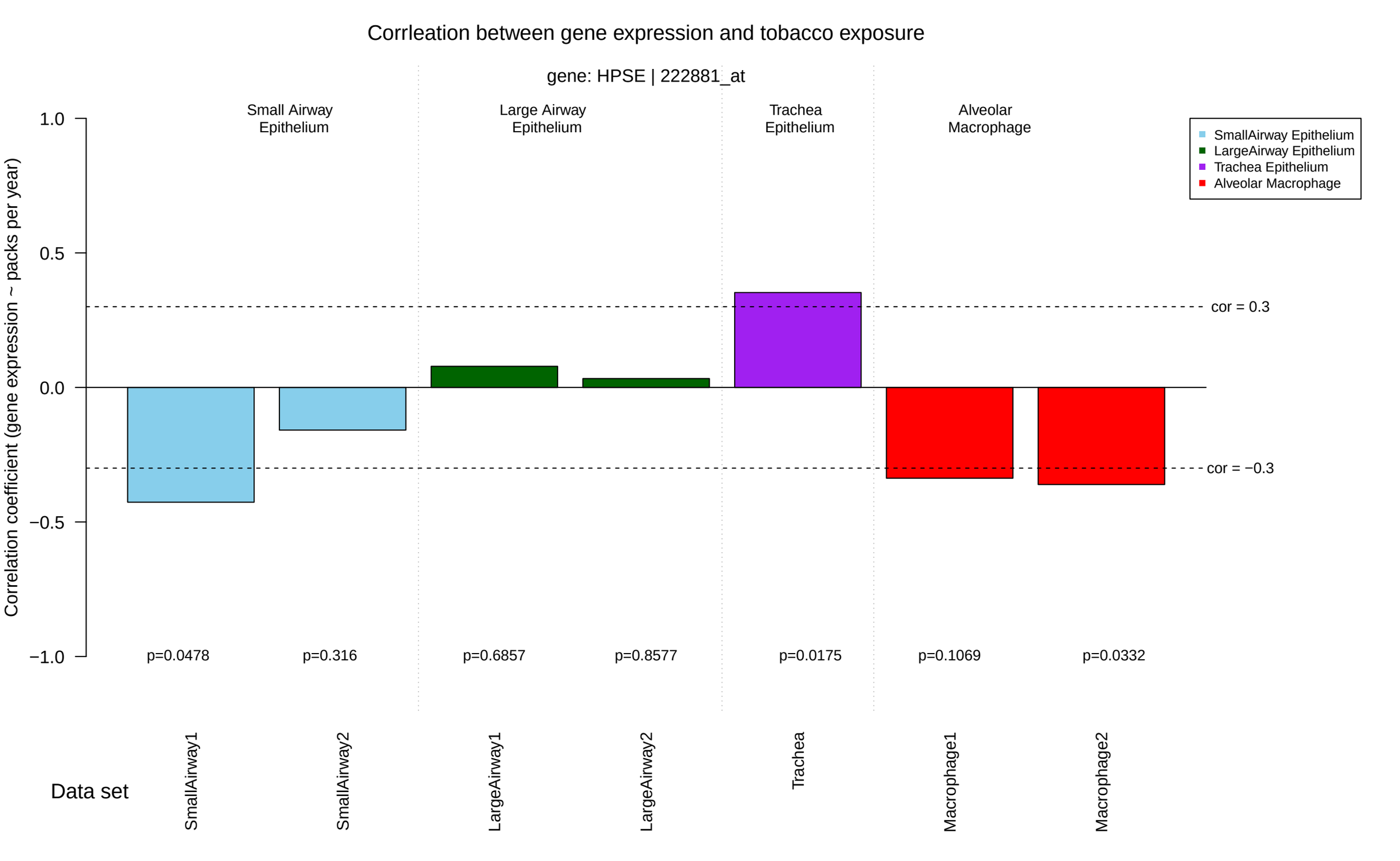

set: LargeAirway1 (p=0.2574)
HPSE | 222881_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9325)
HPSE | 222881_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0602)
HPSE | 222881_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1251)
HPSE | 222881_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1795)
HPSE | 222881_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HPSE | 222881_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0478
p=0.316
p=0.6857
p=0.8577
p=0.0175
p=0.1069
p=0.0332
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2